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Government support for SMEs in the Fintech Era: Enhancing access to finance, survival, and performance☆

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ABSTRACT

Small and medium-sized enterprises (SMEs) are pivotal to global economic growth and stability but often face considerable barriers in accessing finance for growth and survival. Traditional financial support mechanisms present complexities and challenges, limiting the prosperity of SMEs. This systematic literature review (SLR), underpinned by bibliometric analysis, investigates how financial technology (Fintech) can help overcome these challenges. By analyzing 54 empirical studies, this review examines both the direct and synergistic effects of government support and Fintech innovations on SME financing. Our findings reveal a synergistic relationship between government initiatives and Fintech solutions, proposing transformative strategies that depart from traditional approaches that often separate them. Within the context of recent global economic and regulatory shifts, our study underscores the crucial role of policymakers in creating an enabling environment for SME growth through Fintech. This review addresses a key gap in the literature and provides a foundation for future research on the dynamic relationship between government support and Fintech. It emphasizes their combined capacity to transform SME financial empowerment and contribute to sustainable economic development.

1. Introduction

Small and medium-sized enterprises (SMEs) are pivotal in promoting economic progress, forming the foundation of both advanced and emerging economies (Muriithi, 2017). Representing around 90% of businesses worldwide, SMEs significantly contribute to employment and play a critical role in economic dynamism—particularly in emerging markets, where they are substantial contributors to national income (World Bank, 2019).

Despite their significance, SMEs frequently encounter financial constraints, including limited access to capital and finance, leading to failure, bankruptcy, and liquidation (Bandara et al., 2020; Maabreh, 2024). These financial challenges are impaired by issues such as agency costs and the absence of collateral, which complicate the decision-making process for extending finance to these enterprises (Agu et al., 2024; Martinez et al., 2022).

Governments have traditionally intervened to mitigate these challenges (Doh & Kim, 2014; Mavrodieva et al., 2019) by providing essential support in areas such as finance access, skill enhancement, and regulatory framework development (Agu et al., 2024; Ogujiuba et al., 2022). Improving the availability of credit information stands as a key government intervention aimed at reducing informational asymmetries. This measure has been shown to enhance financial inclusion for SMEs, thereby facilitating their access to necessary financing (OECD, 2020).

However, the effectiveness of government support for SMEs is subject to ongoing debate. While Anwar and Li (2021) and Supari and Anton (2022) reported enhanced access to finance, skill development, and improved SME performance through government-backed loan guarantees and subsidies, contrasting findings from Bomani et al. (2022) and Guan and Yam (2015) questioned the efficacy of these interventions, especially in regions where government support has failed to meet the specific needs of SMEs. Choi and Lee (2020) and Pergelova and Angulo-Ruiz (2014) similarly found that, in some cases, government financial support leads to dependency rather than fostering innovation and competitiveness. This inconsistency underscores a considerable research gap, highlighting the limitations of traditional financial support systems in adapting to the evolving needs of SMEs in the digital economy (Feng et al., 2023; Huang, 2022).

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The rise of Financial Technology (Fintech) companies represents a pivotal shift in financial services. Following Gomber et al. (2018), Hanafizadeh and Gerami Amin (2023), Jayesh et al. (2024), and Phani et al. (2023), Fintech refers to the innovative use of digital platforms and advanced technologies, such as machine learning, artificial intelligence (AI), mobile applications, and blockchain, to deliver personalized financial services, improve financial processes, and enhance access to finance—particularly for SMEs. By offering alternative financial solutions and streamlining access, Fintech enables new business models such as peer-to-peer (P2P) lending, mobile applications, and crowdfunding, which reduce barriers to financial access for SMEs (Abbasi, Alam, Du, & Huynh, 2021; Gomber et al., 2018).

Fintech's transformative potential spans numerous banking service domains, revolutionizing traditional financial operations and offering innovative solutions for underserved SMEs (Gomber et al., 2018). Technologies such as AI and machine learning have particularly impacted credit scoring by analyzing alternative data sources, thus allowing SMEs with limited credit histories to access financing and significantly improving support over traditional financial mechanisms (Gomber et al., 2018; Jayesh et al., 2024). In addition to these financial solutions, the growing emphasis on sustainable digital transformation highlights the need for comprehensive roadmaps that guide SMEs through this evolving landscape (Pinto Mick et al., 2024). Blockchain technology provides efficient financial tools, including decentralized finance platforms and smart contracts, which help address credit gaps and improve transaction efficiency for SMEs (Phani et al., 2023). Furthermore, blockchain enhances trust through an immutable ledger, reducing fraud risk, particularly in regions with weaker regulatory environments (Javid et al., 2023). Technological advancements, including mobile applications, open banking, and robo-advisors, have significantly bolstered SME financial inclusion by offering essential services, personalized financial planning tools, and tailored risk management solutions (Gomber et al., 2018; Jayesh et al., 2024). Further, Fintech innovations, such as big-data integration, allow financial institutions to offer highly personalized financial products that cater to the unique needs of SMEs (Acciarini et al., 2023).

Fintech's tailored solutions have been instrumental, particularly in addressing global challenges such as the COVID-19 pandemic, showcasing their pivotal role (Sanga & Aziakpono, 2023; Y. Zhang et al., 2023). These advancements have bridged long-standing disparities in SME financing by providing more accurate risk assessments and including marginalized SMEs. Building on traditional efforts to reduce information gaps, Fintech's innovative toolbox facilitates a more efficient exchange of data between SMEs and financial institutions, effectively addressing these barriers (Brei et al., 2020).

In emerging and less-developed markets, where traditional banking resources and infrastructure are limited, Fintech innovations such as P2P lending and digital payments offer accessible alternatives that promote financial inclusion and resource efficiency. These solutions help SMEs overcome geographical and cost barriers, enabling sustainable and circular business models (Barroso & Laborda, 2022; Giglio, 2021; Maabreh, 2024; Rao et al., 2021; Sanga & Aziakpono, 2023). However, realizing Fintech's full potential in regions such as the South Caucasus and Baltic states requires improvements in digital infrastructure and supportive legislation (Charaia et al., 2021). Despite these advances, challenges such as regulatory hurdles, rapid technological evolution, and potential market disruptions continue to be prevalent (Arner et al., 2015). As Fintech platforms handle increasingly sensitive financial data, concerns over data privacy and cybersecurity are critical across all markets. Without strong cybersecurity protections, SMEs may hesitate to adopt digital financial services, limiting Fintech's potential to expand financial access. Safeguarding financial data against threats such as data breaches, phishing, and ransomware is essential for building trust in Fintech platforms (Hanafizadeh & Gerami Amin, 2023; Javaheri et al., 2024).

In some cases, government interventions may spur technological progress but still fail to meet the specific needs of SMEs, particularly in urban areas (Bomani et al., 2022; Vu & Tran, 2021).

Addressing these challenges necessitates a collaborative approach, blending the agility of Fintech with the supportive frameworks of government policies. Recent studies underscore the critical need for a nuanced understanding of Fintech's impact on the SME sector and its growing importance, particularly in shaping sustainability strategies and practices (Ellili, 2023).

Governments and regulators play a critical role in balancing the need for financial accessibility with investor protection. Regulatory sandboxes in countries such as the UK and Singapore provide controlled environments for testing Fintech innovations, ensuring both market stability and technological progress (Chen, 2019). Moreover, Sheng (2021) emphasized that government collaboration with Fintech in regulatory frameworks crucially ensures adequate credit support from banks and financial institutions to SMEs in China. Government-led initiatives, such as China's digital finance programs and India's Digital India Movement, underscore the importance of aligning Fintech innovations with government efforts to improve SME financing (Feng et al., 2023; Gupta et al., 2022). Such initiatives highlight the potential of a coordinated approach, where regulatory frameworks foster technological advancements, expanding financial inclusion for SMEs, particularly in regions underserved by traditional banking services. (Huang, 2022; Maiti, 2018). Governments can foster scalable and sustainable approaches to financial inclusion for SMEs by encouraging collaboration between traditional financial institutions and Fintech firms. Collevecchio et al. (2024) emphasized that partnerships between Fintech and established banks can drive innovation and improve the financial ecosystem. However, while these partnerships highlight the role of private-sector collaboration, they do not address how government frameworks can facilitate and scale such collaborations. Our study extends this by exploring how government policies can support and promote these partnerships to create scalable solutions for SME financing.

Additionally, Acciarini et al. (2023) highlighted the transformative potential of big data in personalizing financial services and innovating business models for SMEs. However, their work focused on the business-level innovations driven by big data and did not explore the role of government policies in ensuring broader adoption and scaling of these technologies. Our study addresses this gap by investigating how government policies can establish regulatory frameworks to promote the widespread use of big data and other Fintech innovations to enhance SME financial inclusion across diverse economic contexts, with particular consideration for emerging markets.

By supporting such collaborative frameworks at a national level, governments can enhance SME financing and move beyond traditional direct support mechanisms. Flexible regulatory frameworks are crucial for supporting Fintech-driven innovation and security, especially in cross-border contexts (Hanafizadeh & Gerami Amin, 2023). Governments that actively foster Fintech adoption can drive SME growth by expanding access to digital financial services. Collaboration between Fintech firms and traditional financial institutions bridges the gap between innovation and traditional banking services, fostering competitiveness and customer-focused growth in the financial sector (Barroso & Laborda, 2022).

While previous systematic literature reviews (SLRs), such as those by Sanga and Aziakpono (2023) and Rao et al. (2021), have focused on the general relationship between Fintech and SME financing, our study builds on this foundation by exploring how governments facilitate Fintech adoption to enhance SME financial inclusion. These reviews highlight Fintech's role in improving SME financing but often lack in-depth analysis of how government interventions interact with these innovations.

By integrating a government support dimension, our study provides a fresh, comprehensive strategy for addressing financial barriers that neither Fintech nor government support could fully resolve

independently.

In a landscape marked by volatile markets and rapid technological advancements (Troise et al., 2022), this combined influence of government frameworks and Fintech innovations is crucial in enhancing SMEs' financial inclusion and fostering an inclusive economic future.

The timely nature of this inquiry is underscored by recent shifts in the global economy and regulatory landscapes (Gherghina et al., 2020), necessitating a re-evaluation of how policy and technology converge to support SME growth. As Fintech continues to evolve, and governments worldwide seek to modernize their support mechanisms for SMEs, this study is directly relevant to both academic discourse and the practical challenges faced by policymakers. The integration of Fintech innovations with government policies presents a critical opportunity to improve financial access for SMEs, particularly amid ongoing digital transformation and the long-term economic shifts accelerated by the COVID-19 pandemic. This paper addresses these timely issues, offering insights that can guide both immediate policy decisions and future research directions.

Acknowledging the interconnectedness of modern economic ecosystems, as Gherghina et al. (2020) also observed, and addressing the gap in previous research, which often treats government interventions and Fintech solutions separately, this SLR explores how governments can effectively integrate and scale Fintech innovations to enhance SMEs' access to finance. By synthesizing findings from a diverse range of empirical studies, we aim to provide nuanced insights into optimizing government policies in the Fintech era to bolster SME financial access and growth.

We expect our findings to contribute to a broad set of initiatives encouraging innovation and Fintech adoption among SMEs, enriching the academic discourse, and guiding policymakers through digital financial transformation. Additionally, our exploration of the strategic government frameworks and impact of Fintech's rise aims to underscore the importance of regulatory frameworks and technological advancements in adopting financial inclusivity and stimulating economic advancement.

To guide our exploration of the intricate relationship between government policies, Fintech companies, and SMEs, we formulate the following research questions (RQ):

RQ 1: What are the identified limitations and gaps in the literature regarding the financial support provided to SMEs by governments and Fintech when considered separately, and how do these challenges impact SMEs' access to finance and growth?

RQ 2: How do government initiatives, in collaboration with Fintech, provide alternatives to traditional financial support for SMEs, and what challenges impact SMEs' financial access and growth in this context?

RQ 3: Considering the insights from existing research and the identified gaps, what future research directions can be proposed to enhance the synergy between government support and Fintech for the benefit of SMEs?

With these questions as our foundation, we proceed as follows: Section 2 details our methodology, providing a roadmap for our investigation. Section 3 combines the results and discussion, offering a comprehensive analysis of our findings. Section 4 outlines potential future research directions, shedding light on unexplored areas that could further inform policymaking. Section 5 provides policy recommendations. Finally, Section 6 summarizes our findings and discusses their implications.

2. Methodology

In this study, we used an SLR methodology guided by PRISMA guidelines (Moher et al., 2009) to explore how previous literature has handled the impact of government policies and Fintech on SME finance access. PRISMA's structured approach ensured transparency and reproducibility throughout the review process, including clear documentation of study identification, screening, eligibility, and inclusion stages. This method is supported by Pahlevan-Sharif et al. (2019), who advocated for adopting PRISMA in the social sciences to enhance transparency, reliability, and validity despite its origins in medical research. Literature reviews are essential tools for assessing the state of knowledge in a field, identifying gaps, and suggesting potential future developments (Grant & Booth, 2009).

Our SLR serves as a potent analytical instrument, distilling the existing literature to bridge crucial knowledge gaps and provide synthesized insights for policymakers, financial bodies, and SMEs. Benchmarks set by Verma et al. (2023) and Sanga and Aziakpono (2023) demonstrated the efficacy of SLRs in meticulously aggregating and evaluating scholarly work within the Fintech realm. The choice to employ an SLR was made with the aim to compile a comprehensive dataset, especially in areas where direct studies are sparse, allowing for a broad synthesis of available research, thereby uncovering trends and identifying research gaps. These tools enabled us to address our research questions by identifying key limitations in government financial support (RQ1), evaluating Fintech's role as an alternative mechanism (RQ2), and uncovering emerging trends and gaps in the literature (RQ3).

To enhance the review process, we used Covidence, a widely used web-based software, to streamline study selection and data extraction (Covidence, 2022). Several studies have validated the efficiency and accuracy of Covidence (Castelein et al., 2020), supporting the rigor of our methodology.

For bibliometric analysis, we utilized Bibliometrix, an R package that enables comprehensive bibliometric and scientific analysis (Aria & Cuccurullo, 2017). Its companion web app, Biblioshiny, provides a user-friendly interface, which allowed us to visualize and explore trends effectively (Donthu et al., 2021; Thangavel & Chandra, 2023). Agostini et al. (2023) demonstrated the effectiveness of bibliometric analysis to uncover research gaps and trends in strategic flexibility, highlighting the value of such methods in addressing key research questions. These methodological insights informed our analysis and contributed to addressing RQ3, focusing on identifying emerging trends and gaps in the literature.

2.1. Database selection and search strategy

To underpin our review, we selected the Scopus and Web of Science databases for their comprehensive coverage and detailed indexing capabilities, ensuring the inclusion of a wide array of relevant studies and facilitating a thorough analysis of the literature within a given field (Calvo Rubio & Ufarte Ruiz, 2021). These databases are particularly valuable for conducting systematic reviews that span interdisciplinary research—a key factor for addressing the complexity of Fintech, government policies, and SME financial access. Scopus and Web of Science are renowned for their high-quality, peer-reviewed content, and extensive coverage of journal articles, conference papers, and books. This breadth is particularly important for ensuring that we capture all relevant literature on the intersection of Fintech and SME financing, especially given the global and interdisciplinary nature of this topic. This approach, grounded in prior scholarly work, is crucial for the relevancy and accuracy of our search outcomes and aligns with recommendations for capturing the most pertinent studies within our thematic scope (Mohamed Shaffril et al., 2021).

Following Pahlevan-Sharif et al. (2019) and Pickering and Byrne (2014), we used predefined keywords such as "SMEs," "Fintech," "government support," and "access to finance," which are central to our study's domain, in both databases. These were combined with Boolean operators "AND" and "OR," enhancing the comprehensiveness and thoroughness of the search—a strategy further analyzed using the author's keyword, as illustrated in recent methodological studies.

The search set temporal boundaries from 2009 to 2023, marking a pivotal shift in financial paradigms and marking the ascendancy of Fintech as an important player in SME financing (Arner et al., 2015). By selecting English-language articles, we aimed to maintain consistency

and broaden the academic reach of the findings within global discourse (Linares-Espinós et al., 2018).

This search and selection process effectively captured core themes relevant to our research objectives, addressing critical research gaps and deepening the understanding of the interplay between government support and Fintech innovations, particularly in relation to their impact on SME financial access.

2.2. Study selection

In shaping the scope of our SLR, we adhered strictly to the PRISMA guidelines, guided by established methodological principles from Moher et al. (2009), focusing on studies that directly informed our research questions and objectives (see Fig. 1). Initial searches in the Scopus and Web of Science databases yielded 385 studies. After removing 62 duplicates, we proceeded with a dataset of 323 unique entries. We then proceeded with a rigorous examination of titles and abstracts, which allowed us to distill the collection further into 225 studies aligning with the core themes of our research: SMEs, Fintech, government support, and access to finance.

We applied subsequent scrutiny to this subset to ensure alignment with our research objectives. Following Castelein et al. (2020), we utilized search terms and expert consultation to refine their study selection, applying a rigorous screening process to ensure relevance and quality. This led to excluding 151 studies that lacked direct relevance or exhibited insufficient empirical depth, along with 20 studies that were excluded owing to accessibility issues. This process distilled the pool to 54 empirical studies integrals to our review, each selected for its substantial insights into the intersection of SME financing, governmental interventions, and technological advancements.

2.3. Bibliometric and thematic analysis

Building on the literature-selection process, we advanced to a dual-strategy analysis to capture the full spectrum of global insights. Through thematic analysis, we decoded the varied contexts within which government policies and Fintech innovations operate, understanding their influence on the financial operations of SMEs across diverse geographies. This crucial exploration, grounded in the work of Donthu et al. (2021), ensured that our insights were expansive and reflective of the global research landscape.

Our thematic and bibliometric analyses allowed us to separately evaluate the contributions of government support and Fintech interventions to SME finance access, ensuring a clear distinction between these two factors across both developed and emerging markets. However, owing to the distinct challenges in emerging markets, we especially emphasized studies addressing SME financial access in these regions.

Biblioshiny, an extension of Bibliometrix, facilitated visual mapping of research trends, citation networks, and keyword co-occurrences across fields such as technology adoption, innovation, and SME financing (Aria & Cuccurullo, 2017; Donthu et al., 2021; Thangavel & Chandra, 2023). This quantitative approach enabled us to trace the evolution of key topics, providing a transparent and structured review of academic discourse.

To further capture the temporal development of these themes, we intend to determine how Fintech and regulatory frameworks internationally have influenced the discourse within this field. A chronological assessment of academic publications is planned to add further context to the thematic evolution, thereby offering a longitudinal view of how important global events and technological progress have shaped academic and policy discussions.

2.4. Analytical synthesis of the literature

Moving forward, our research process involved critically analyzing each article, assessing methodologies, results, and discussions from the selected studies, and comparing different scholarly perspectives to highlight consensus areas and divergent views. Our synthesis aimed to deepen the collective understanding of the complex interplay between

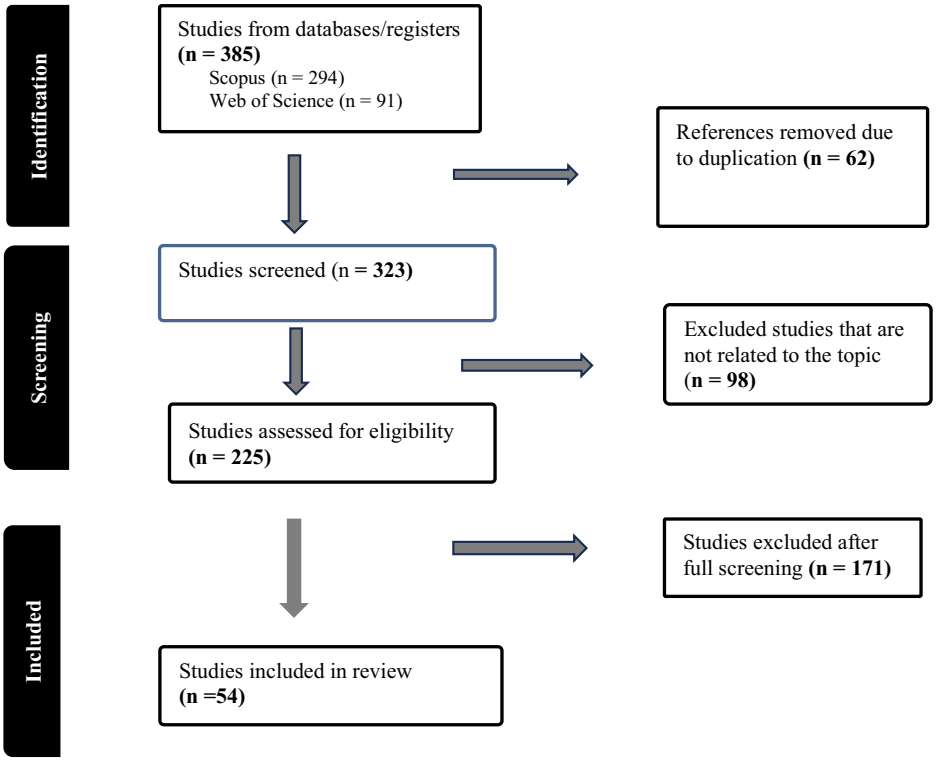


Fig. 1. The PRISMA flow diagram. Source: Adapted from Moher et al. (2009)

government policy, Fintech, and SME financial access. Following the framework of Mohamed Shaffril et al. (2021), we crafted a holistic view that captures the various dimensions of SME financial empowerment.

We analyzed each study for its individual contribution and its part in the collective academic dialogue; this allowed us to identify trends, gaps, and cumulative wisdom within this domain. In addition to structuring the synthesis around scholarly perspectives, we assessed each study for its contribution to the evolving discourse on SME financial access. This approach allowed for a comparative analysis that highlighted commonalities and divergences in government policies and Fintech innovations across different geographic contexts. This structured analysis facilitated the identification of patterns that answer RQ1, RQ2, and RQ3, reinforcing the importance of a holistic view of SME financial empowerment.

Our findings illustrated the transformative potential of government support in concert with Fintech innovations, providing a rich tapestry of insights into SME financial strategies and outcomes.

3. Result and discussion

3.1. Bibliometric analysis

Our review, spanning from 2009 to 2023, encompasses not only the direct interactions between government financial support and Fintech innovations but also the broader implications for enhancing SME financial operations. Using bibliometric analysis, we identified key trends and gaps in the literature and mapped the intellectual landscape of Fintech and SME finance, with Biblioshiny aiding in visualizing and examining data in detail. The findings from our bibliometric analysis set the stage for a more in-depth discussion of the implications for small businesses, government bodies, and the Fintech industry.

The data presented in Table 1 underscore a robust increase in scholarly interest surrounding the integration of government support with Fintech in SME financing, evidenced by an annual publication growth rate of 13.65%. This trend not only reflects the sector's expanding academic importance, but it also highlights its practical significance in the modern economic landscape.

This growing academic focus underscores the relevance of our research questions, particularly the exploration of limitations and gaps in government and Fintech financial support for SMEs when considered separately (RQ1) and the potential of Fintech as an alternative to traditional government mechanisms (RQ2). With an average document age of 4.24 years, the field is characterized by ongoing dialogue and current developments. The considerable average of 23.52 citations per document indicates that the research outputs are impactful and well engaged within the academic community.

Table 1
Bibliometric overview of literature on government support and fintech in SME finance.

Description	Results
Timespan	2009:2023
Sources (Journals, Books, etc.)	48
Documents	54
Annual Growth Rate %	13.65
Document Average Age	4.24
Average Citations per Doc	23.52
References	2997
Document Contents	
Author's Keywords (DE)	216
Authors	158
Authors of Single-Authored Docs	7
Authors Collaboration	
Single-Authored Docs	8
Co-Authors per Doc	3.07
International Co-Authorships %	29.63

Source: Biblioshiny.

The diversity reflected in the 48 sources and 158 authors speaks to a multidisciplinary and collaborative approach to research, with nearly 30% international co-authorships further illustrating the global relevance of this topic. The distribution of 216 authors' keywords depicts a rich and complex thematic landscape, indicative of the multifaceted nature of this research area. This bibliometric snapshot provides a solid base for the ensuing qualitative analysis, setting the groundwork for a comprehensive narrative that will explore the current state and projected trajectory of the interplay between government policy, Fintech, and SME finance, paving the way for future research that aims to enhance the integration of government support with Fintech innovations (RQ3).

3.1.1. Annual scientific production

Fig. 2 shows a noticeable increase in the number of articles focusing on government support and Fintech within SME financing, which highlights a growing academic and practical interest in these areas. This trend aligns with discussions on the annual growth rate and average citations per document, emphasizing the field's expanding influence. The dataset, compiled from both Scopus (76.36%) and Web of Science (23.64%) before filtering, reflects contributions from diverse regions and disciplines, underscoring the global relevance and interdisciplinary nature of this research topic.

The graph shows a substantial peak in 2021, underscoring an acute scholarly response to global economic challenges, such as the COVID-19 pandemic, and the rapid acceleration of digital solutions within the SME financing sphere (Karim et al., 2022; Y. Zhang et al., 2023). As SMEs sought new avenues for survival, governments and Fintech companies responded by introducing interventions such as emergency financial packages, digital credit solutions, and regulatory flexibilities, creating new research opportunities that reflect the Fintech landscape' rapid evolution. This period marks a critical confluence of technological innovation and policy adaptation, reflecting the resilience and agility of SMEs in leveraging Fintech solutions amidst adversity (Semenyuta et al., 2019).

The fluctuations in the graph—including the dips and rises—may be attributed to various factors, including economic downturns, shifts in government policies, or substantial technological breakthroughs that periodically boosted research interest (Hua & Huang, 2021; Sybirianska et al., 2018). This temporal snapshot captures the dynamism of the field and suggests that, while immediate interest may vary, the underlying importance of these research areas remains highly relevant. Such trend analysis provides a historical overview and serves as a barometer for future research trajectories. It underscores the importance of continued scholarly inquiry into how government support, coupled with Fintech, contributes to the financial resilience of SMEs (Nguyen, Van, et al., 2018).

Future research could focus on examining the long-term impacts of these interventions—particularly in emerging markets, where innovative financial solutions are most needed. Additionally, exploring the potential of government-Fintech partnerships in fostering sustainable business practices and inclusive economic growth could offer valuable insights for policymakers and practitioners alike. These insights can inform the development of policies aimed at harnessing the collaborative power of these sectors to support SMEs amidst ongoing digital transformations in the economy (Huang, 2022).

3.1.2. Country production analysis

The analysis of scientific production across various countries, as depicted in Fig. 3, reveals that China leads with approximately 19.31% of the total research output in the domain of SME financing via Fintech and government support. China's proactive policies, large market size, and rapid adoption of Fintech innovations have positioned it as a global leader in this field, which reflects its strategic focus on technology-

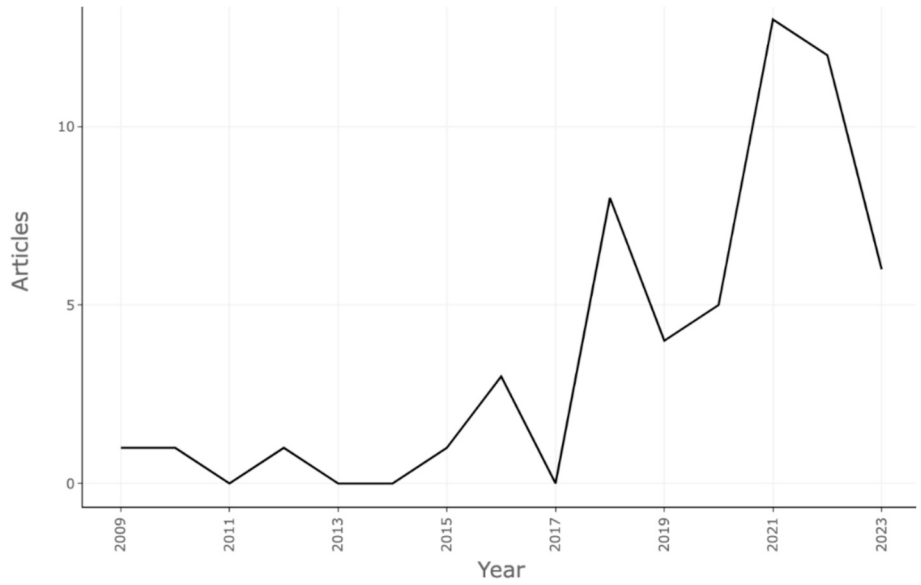


Fig. 2. Publication trends over time.
Source: Biblioshiny.

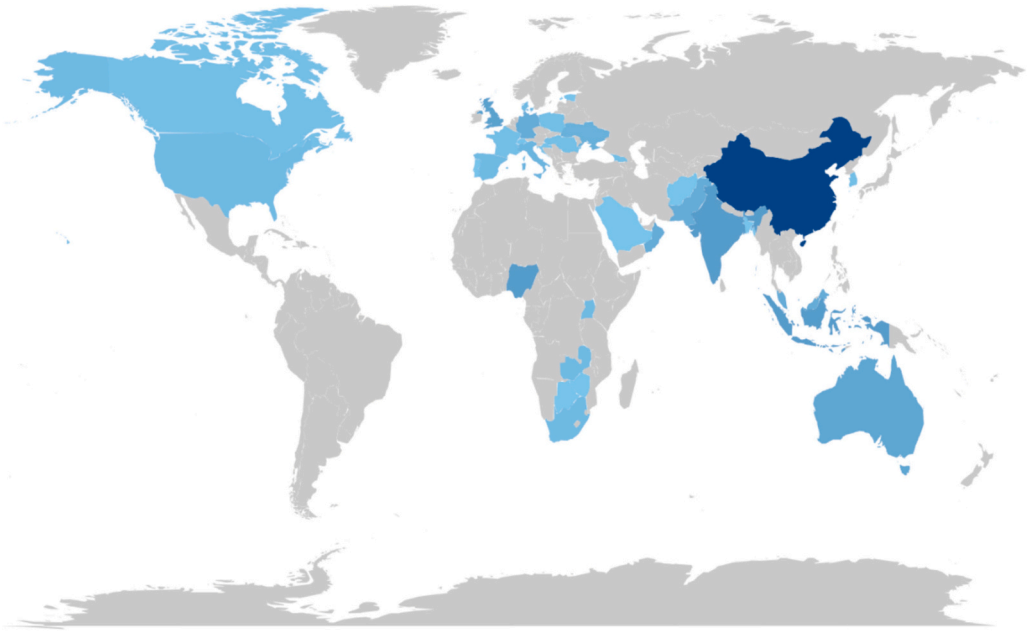


Fig. 3. Country scientific production.
Legend: Light blue – less production, Dark blue – more production. Source: Biblioshiny.

driven growth and economic priorities (Hua & Huang, 2021; Huang, 2022).

Countries such as India, Indonesia, and Nigeria each represent 6% of the total publications. Their research contributions reflect a growing recognition of integrating Fintech with traditional financial systems to enhance SME access to finance (Hua & Huang, 2021; Lawal et al., 2018; Rosavina et al., 2019). For instance, Nigeria's focus on Fintech solutions to overcome infrastructural challenges provides a model for other emerging markets facing similar constraints (Lawal et al., 2018). Meanwhile, India's "Digital India" initiative and Indonesia's digital financial inclusion programs align with a broader regional trend of using Fintech to drive economic development and expand financial access in underserved areas (Gupta et al., 2022; Suharyati & Utami, 2022).

The UK, which contributes 8% of the publications, is also a notable player in the Fintech-SME discourse. Its pioneering work with regulatory sandboxes encourages Fintech innovation while ensuring regulatory compliance, positioning the UK as a key influencer in global Fintech ecosystems. This proactive approach makes the UK a valuable contributor to the Fintech-SME research landscape.

In contrast, some advanced economies, including the USA (2%), Italy (4%), and Spain (3%), demonstrate low research outputs in this field despite their established roles in global Fintech and SME finance ecosystems. This discrepancy highlights potential areas for increased research focus in these regions, particularly given their influence on financial innovation.

Collectively, these contributions underscore the global relevance of Fintech innovations and government support strategies, offering diverse insights for policymakers, financial institutions, and business leaders

operating across different regulatory and technological contexts (Karim et al., 2022;Semenyuta et al., 2019; Suharyati & Utami, 2022).

Furthermore, this diversity underscores the importance of cross-cultural research in evaluating how local economic conditions, cultural nuances, and regulatory frameworks influence the effectiveness of Fintech solutions and government policies. Such insights from cross-cultural research are crucial for developing more nuanced and effective financial support systems that can cater to the diverse needs of SMEs worldwide.

3.1.3. Collaboration networks

The map of the author's collaboration network map (Fig. 4) reveals robust academic networks and shared endeavors that fuel the field's progress. Nodes of varying sizes illuminate the researchers' individual contributions, with larger nodes such as that of "Abbasi K." indicating authors who have had a substantial impact within the scholarly community through extensive publications and high citation counts. This node's prominence highlights extensive involvement and points to a broader trend toward interdisciplinary collaboration, which is crucial for spurring innovation in SME financing strategies (Abbasi, Alam, Du, & Huynh, 2021). Such collaborations often span across fields such as economic policy, technological innovation, and business management, enhancing the richness and applicability of research (Karim et al., 2022; Pizzi et al., 2021).

However, the map also shows that collaboration networks in this area are sparse and dispersed, possibly owing to the nascent nature of focused research on Fintech and government support within SME financing. This sparsity not only suggests challenges in knowledge dissemination, but it also highlights the potential barriers, such as geographic, linguistic, or disciplinary divides, preventing the full utilization of collective expertise. To overcome this, research institutions could leverage digital platforms for more frequent, real-time collaboration and foster open-access publications, ensuring that insights from diverse regions are more widely accessible. Moreover, stronger institutional support for cross-border academic exchanges and research grants could serve as powerful catalysts for building these networks. This observation underscores the need for more robust collaboration ties that extend from regional to global perspectives (Semenyuta et al., 2019; Suharyati & Utami, 2022). Advancements in the field are not solely the result of isolated efforts but emerge through synthesizing diverse insights and expertise (Hua & Huang, 2021).

A visualization of these collaboration networks highlights the value of fostering a collaborative research culture, which is pivotal in developing comprehensive solutions to the multifaceted challenges of SME financing (Karim et al., 2022). As scholars come together to share findings and build upon each other's work, they collectively contribute to developing more effective policies and refining Fintech tools that empower SMEs globally (Nguyen, Tran, & Do, 2018). Expanding these networks could significantly enhance the scope and impact of research, addressing both current and emerging challenges in the sector (Pizzi et al., 2021).

Fig. 5 presents a co-occurrence network that visually delineates two prominent clusters, colored red and blue, each symbolizing distinct research trends within the domain of SME financing. This representation highlights the field's multifaceted nature and the diverse approaches to studying SME financial empowerment.

At the heart of the network, the red cluster, which prominently features "Fintech" at its core, is directly connected to "SMEs." This illustrates the critical role of Fintech innovations, encompassing tools that expand SMEs' access to flexible and sustainable financing options, such as participative loans and long-term financing (Bertoni et al., 2019; Karim et al., 2022). Additionally, this cluster includes nodes such as "China," "COVID-19," "circular economy," and "government support," illustrating a dynamic interplay between technological advancements and macroeconomic factors.

The presence of "China" and "COVID-19" within this cluster points to

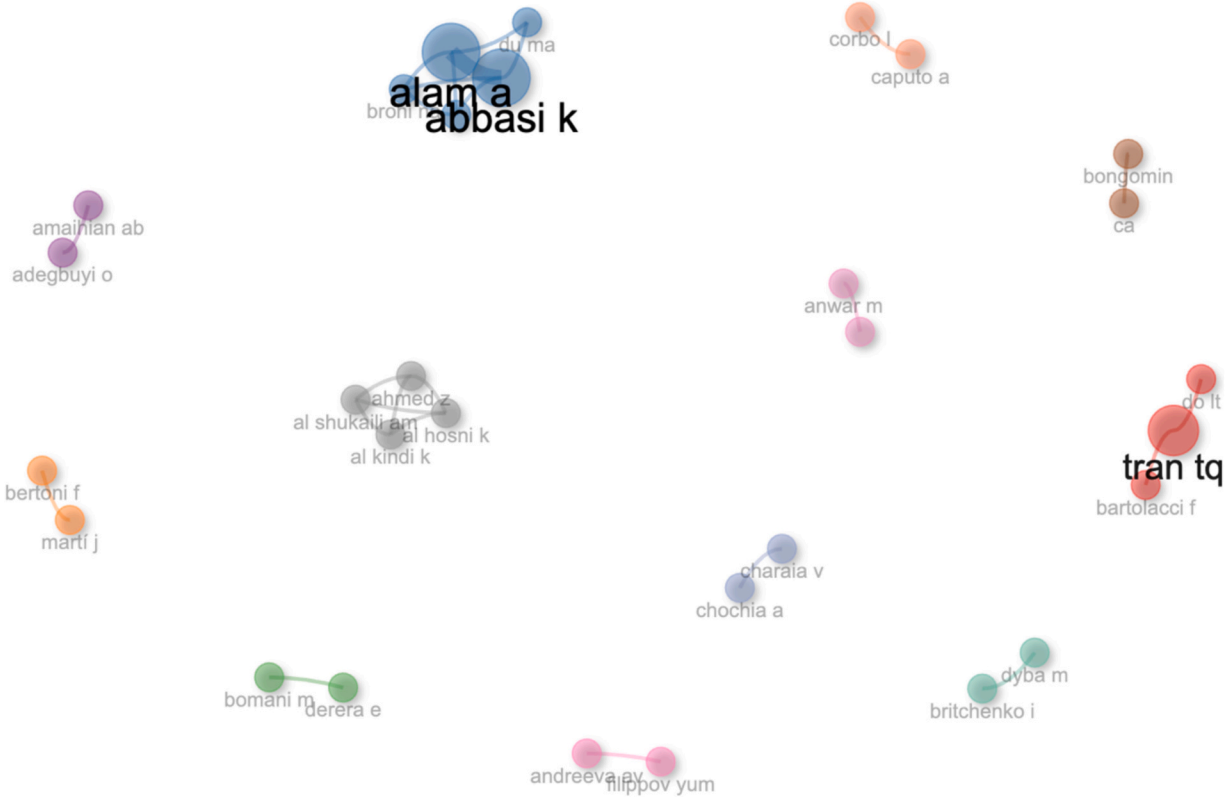


Fig. 4. Author collaboration network map.
Source: Biblioshiny.

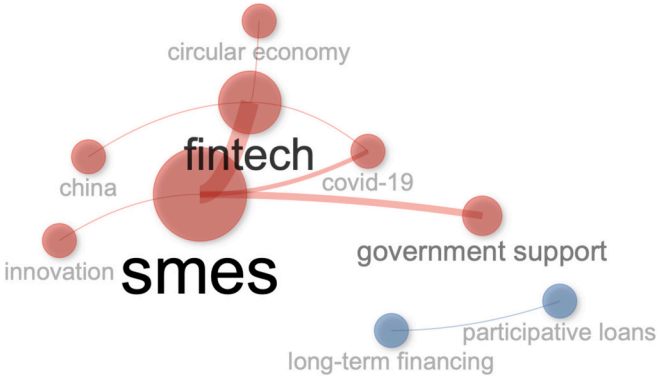


Fig. 5. Co-occurrence network.
Source: Biblioshiny.

the substantial impact of geographic and global health dynamics on the adoption of Fintech solutions and their adaptation in response to crises (Xiang et al., 2021; Y. Zhang et al., 2023). The inclusion of "circular economy" and "innovation" within this cluster also suggests a broader context where Fintech is not only driving financial inclusion but is aligned with sustainable practices. This intersection indicates that Fintech is being leveraged to support more sustainable economic models, such as resource management strategies, which are essential in the shift toward a circular economy. Government support within this cluster plays a pivotal role, highlighting how policy initiatives foster an environment conducive to both technological advancements and sustainability (Hua & Huang, 2021; Karim et al., 2022).

In the blue cluster, terms such as "participative loans" and "long-term financing" emphasize a focus on alternative financing models that cater to SMEs' specific needs. These financing options allow SMEs greater flexibility and stability, providing them with avenues for capital that go beyond traditional banking structures. Participative loans represent a hybrid financing approach that combines elements of debt and equity, offering adaptable repayment options that can be crucial for SMEs in uncertain economic conditions (Bertoni et al., 2019). This cluster highlights the emphasis on sustained financial support structures that ensure SMEs have access to growth capital over the long term (Feng et al., 2023).

The strategic positioning of "government support" between the red and blue clusters suggests its bridging role in connecting technology-driven Fintech solutions with structured, sustainable financing models. This position underscores the importance of regulatory frameworks in harmonizing innovative Fintech solutions with traditional financing structures, thereby supporting SMEs comprehensively. Evolving regulations are essential for maintaining balance, ensuring safety, protecting both SMEs and investors and fostering trust in emerging financial technologies (X. Zhang et al., 2022; Hua & Huang, 2021).

Future research could delve into the comparative impacts of these alternative financing models in different regulatory contexts, examining how countries with varying levels of Fintech adoption integrate these innovations with traditional financial systems. Longitudinal studies on the impact of participative loans and long-term financing models on SME growth and sustainability could provide actionable insights for policymakers aiming to strengthen SME resilience and adaptability (Bertoni et al., 2019; Karim et al., 2022).

By mapping the connections across terms such as "Fintech," "SMEs," "government support," and "long-term financing," Fig. 5 provides a comprehensive overview of the current state of research in SME financing. This visualization emphasizes the need for a nuanced understanding of how financial models, technological advancements, and regulatory policies intersect to shape opportunities and challenges for SMEs across diverse economic contexts (Nguyen, Van, et al., 2018; Pizzi et al., 2021). This highlights the importance of interconnected strategies for resilient SME financing in diverse economic contexts.

3.1.4. *Countries collaboration network*

The global landscape of research collaboration within SME financing, as depicted in Fig. 6, offers valuable insights into the interconnectedness of various countries and regions in advancing knowledge and practices in this domain. The visualization reveals four distinct clusters, each representing different collaboration patterns and highlighting unique contributions to the field.

The blue cluster, comprising the USA, Malaysia, Singapore, and Switzerland, showcases a robust exchange of insights and knowledge among countries with diverse financial ecosystems. This collaboration fosters innovation and cross-pollination of ideas, contributing to developing adaptive financial frameworks for SMEs that incorporate advanced risk-management tools and digital payment systems.

Conversely, the red cluster, led by China and joined by Germany, the United Kingdom, and Pakistan, represents a collaboration among major players in the global financial landscape. China's central position within this cluster underscores its strategic initiatives in Fintech and active support programs for SMEs, while other countries contribute to the exchange of research and best practices in financial technology deployment (Hua & Huang, 2021; Huang, 2022).

The green cluster, consisting of Portugal and Italy, signifies collaboration within European nations, highlighting the importance of regional partnerships in advancing SME financing research. Despite being less dense compared to other clusters, this collaboration fosters knowledge-sharing and mutual learning, contributing to the resilience of SMEs in the European context. Specifically, these partnerships have focused on sustainable financing models that are crucial in the European Union's regulatory environment (Flögel & Beckamp, 2020).

The purple cluster, comprising India and Oman, serves as a bridge between emerging markets and other clusters, emphasizing the role of these countries in connecting disparate research efforts globally. This collaboration facilitates the exchange of ideas and expertise, driving innovation and addressing the unique challenges faced by SMEs in emerging economies, such as access to microfinancing and integration into global trade networks (Feng et al., 2023; Hua & Huang, 2021).

However, the visualization also suggests that in some regions, such as the Middle East and Africa, the network is less dense, which implies room for increased research integration and collaboration. The limited network density may be due to factors such as restricted access to advanced Fintech infrastructure, regulatory challenges, or economic instability. Addressing these barriers through targeted initiatives could enhance these regions' participation in global SME financing discourse. This highlights a substantial opportunity for growth, encouraging a more inclusive research dialogue that extends the reach of Fintech innovation and government support strategies to these emerging markets (Lawal et al., 2018; Suharyati & Utami, 2022).

Thus, the network map not only charts the current state of international collaboration, but it also echoes the need for a broader angle in research efforts. Future research could explore strategies to enhance collaboration in underrepresented regions, such as the Middle East and Africa, through case studies or pilot programs that focus on overcoming local challenges. Additionally, comparative studies between densely and sparsely connected clusters could provide valuable insights into best practices for fostering global SME financial empowerment. It advocates for expanding the academic and practical scope to ensure that the benefits of government support and Fintech innovations are shared more equitably across all regions, fostering a truly global advancement for SME financial empowerment.

3.1.5. *Trend analysis*

The trend analysis graph in Fig. 7 illuminates the evolving priorities within SME financing discourse from 2018 to 2022, tracing the prominence of core concepts such as "government support," "SMEs," and "Fintech." The persistent emphasis on "government support" throughout

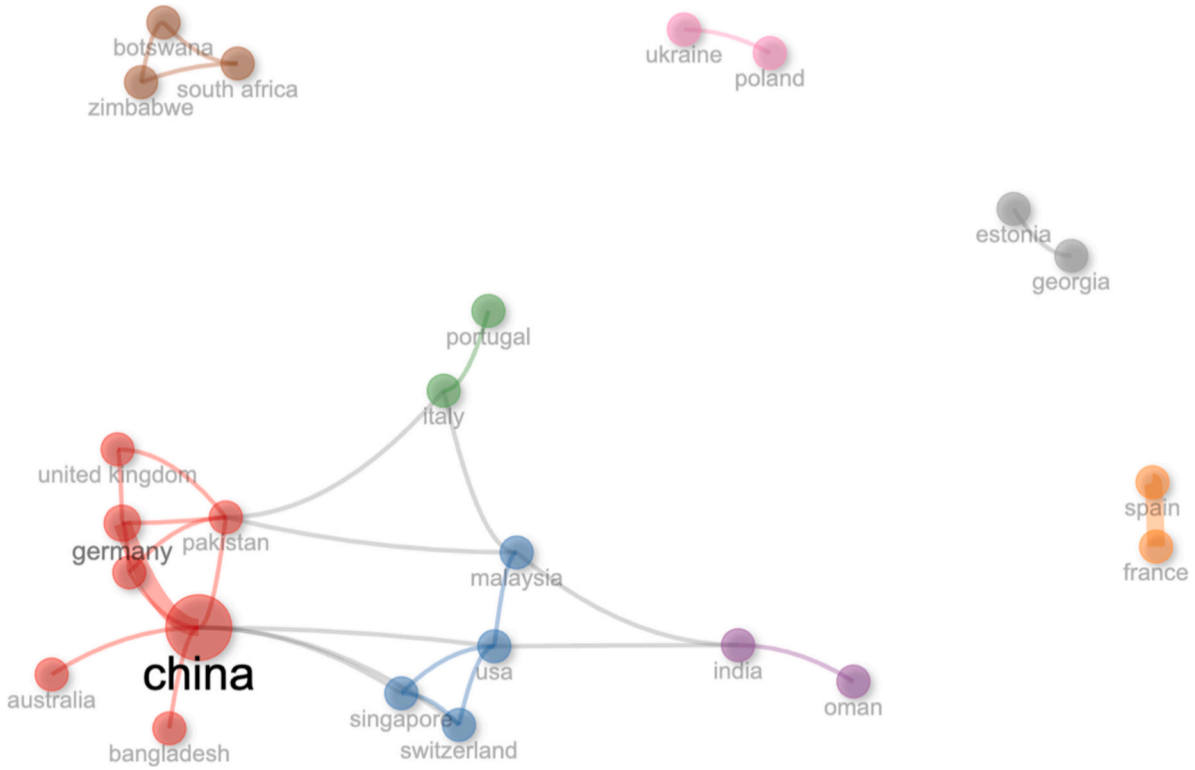


Fig. 6. International collaboration map.
Source: Biblioshiny.

this period aligns with global initiatives to bolster SMEs, revealing the unwavering relevance of policy interventions in various economic conditions (Karim et al., 2022; Nguyen, Van, et al., 2018). "SMEs" continue to be a central focus, which points to their foundational role in economic discourse and highlights their continual impact on the broader financial ecosystems (Siddik et al., 2023).

The graph also shows how "Fintech" has gained traction, particularly highlighted by its increased mention in recent years. This shift represents a growing pivot toward digital solutions in financial ecosystems, likely spurred by both advancing technology and the changing needs of SMEs in a digital age (Huang, 2022; Pizzi et al., 2021). These trends reflect not only shifts within the SME financing domain but also broader global economic and technological transformations. The increasing digitization across industries and the growing emphasis on financial inclusion have likely contributed to the rising prominence of Fintech in recent research.

The discrete but noteworthy emergence of "COVID-19" starting in 2020 underscores the pandemic's catalytic effect on research themes, particularly concerning SME resilience and adaptation strategies. This trend signifies the research community's responsiveness to emergent challenges, showcasing a dedication to exploring the efficacies of policy and technological innovations under crisis conditions (Karim et al., 2022; Y. Zhang et al., 2023). The impact of the pandemic has led to increased discussions on how both government support and Fintech can be leveraged to ensure the survival and growth of SMEs during and beyond global crises (Xiang et al., 2021; Y. Zhang et al., 2023).

This analysis charts the relevant themes over time and provides insights into how the interplay between these areas can guide future research and policymaking. It underscores the need for a dynamic approach to SME financing—one that adaptively integrates technological advances and robust governmental frameworks to meet the evolving challenges and opportunities within the SME sector (Nguyen, Tran, & Do, 2018; Pizzi et al., 2021). Future research should focus on the intersection of government support and Fintech, particularly in the context of crisis response and long-term SME sustainability, ensuring that SMEs are equipped to thrive in an increasingly dynamic global economy.

3.2. *Findings from content analysis*

Our systematic review and bibliometric analysis synthesize these findings, painting a picture of a research community that is both reactive to immediate challenges and proactive in strategic planning. Employing content analysis, we systematically reviewed 54 critical articles to uncover the intricate dynamics among government support, Fintech innovation, and SME financial access. This methodology allowed for a comprehensive examination that extends the traditional understanding of government support, revealing Fintech's unique capacity to facilitate broader financial inclusion for SMEs.

The synthesis from the reviewed literature extends beyond the mere survival of SMEs to their thriving in a competitive and digital economy. Our findings underscore the importance of maintaining a broad, inclusive perspective to ensure that the integration of innovative Fintech solutions and substantial government support continues to drive global SME support. This perspective not only highlights the dynamic and evolving nature of SME financing, but also suggests that the field is ripe for the development of novel analytical tools and frameworks. These tools are essential for capturing the complexity of SME financing and pushing the boundaries of what is currently understood about the interplay between government policy, technological advances, and SME sustainability.

The studies are summarized into three thematic groups, as detailed in Table 2. These groups reflect the shared findings and insights, facilitating a coherent discussion of the interactions between identified areas. The organization of the literature into thematic groups reveals an ongoing evolution in their interrelationships and accentuates the necessity for more nuanced research to understand the intricacies therein. A complex web of interactions was found, showcasing the dualistic

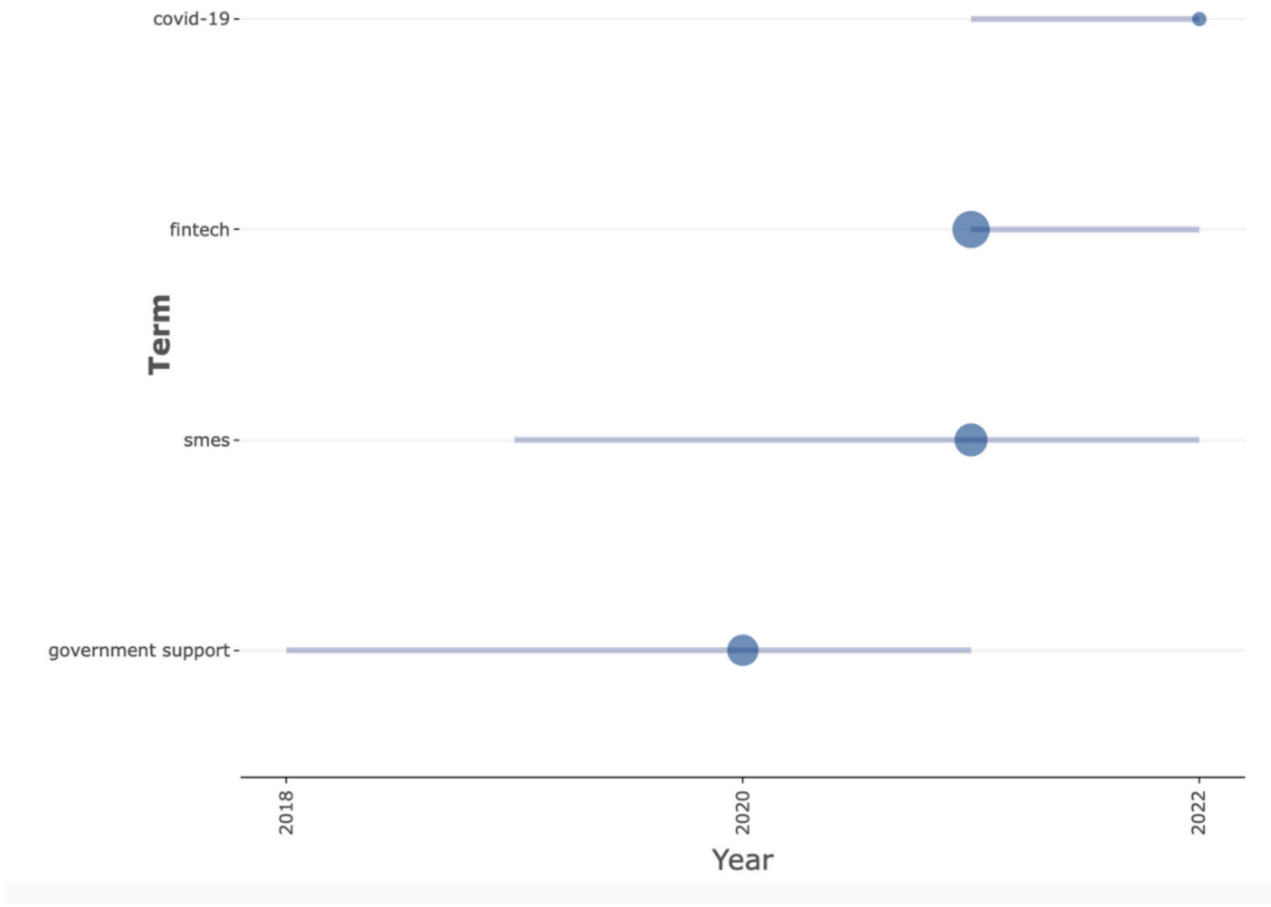


Fig. 7. Trend topics.
Source: Biblioshiny.

nature of government engagement and Fintech innovation in offering both challenges and opportunities for SME financing. Through this synthesis, we establish a baseline for in-depth discourse, poised to further dissect and interpret the core findings.

3.2.1. Sector 1: The traditional government financial support for SMEs

3.2.1.1. Government financial support is effective for SMEs. To effectively address RQ1, this sector delves into the traditional mechanisms of government support for SMEs, pinpointing pivotal roles these supports play and identifying areas that require refinement. Government support for SMEs—a cornerstone of economic policy—plays a pivotal role in facilitating SME growth and competitiveness, as demonstrated in the Omani context, where government-backed firms exhibit enhanced job-creation capabilities (Al Shukaili et al., 2022). In Oman, the job-creation capabilities of government-backed firms do not merely address unemployment but foster an entrepreneurial ecosystem that boosts the broader economy by encouraging innovation. This creates employment and enhances labor market dynamism, a critical factor for sustainable economic development. This positive relationship is indicative of the broader trend where government intervention aids in immediate financial relief and long-term enterprise growth (Anwar & Li, 2021; Bertoni et al., 2019).

However, current government programs often adopt a one-size-fits-all approach, which may not fully support the diverse needs of SMEs across various sectors and stages of growth (Jayeola et al., 2022; Klonowski, 2010). Tailoring government financial support programs based on SME size, industry, and business lifecycle would help maximize their impact and long-term success. Customized support mechanisms would ensure that SMEs with varying financial health and market positioning can equally benefit from these programs, helping address disparities and promote a more equitable distribution of benefits (Msomi & Olarewaju, 2021).

Similarly, stakeholders in Saudi Arabia call for improved stimulus programs, recognizing that current efforts, while beneficial, require fine-tuning to better address post-COVID-19 recovery needs and digital upgrades necessary for SME sustainability (Alharbi, 2023). This need for refinement is echoed across various economies, which highlights the importance of government support in SMEs' adaptation to technological advancements.

In Pakistan, Alkahtani et al. (2020) found a strong positive impact of government financial support on the sustainable competitive performance of SMEs. This underscores the broader significance of government intervention in developing economies and its role in enhancing SMEs' market positioning through networking structures. This sector not only recounts successful interventions across different geographies, but also underscores the crucial need for policies to evolve in response to new economic realities. For instance, findings from Spain (Bertoni et al., 2019) demonstrate how government-supported participative loans significantly contribute to the growth of entrepreneurial ventures in young, small, or high-tech sectors. Similarly, in China, Anwar and Li (2021) argued for the necessity of targeted government incentives to bolster SME performance; this underlines the need for these programs to be dynamic and well suited to rapidly changing market conditions and sector-specific demands.

In post-conflict Uganda, Bongomin et al. (2018) illustrated how government support modifies the relationship between various business factors—such as skills, capital adequacy, and access to finance—and

Table 2
Findings from content analysis on fintech and government support for SME finance access.

Themes	Findings	Authors
C1: The Traditional Government Financial Support. a. Effective for SMEs	Government support is crucial in facilitating easier access to finance for SMEs.	Msomi and Olarewaju (2021), Supari and Anton (2022), and Wang et al. (2021).
	Government financial assistance significantly enhances SME growth.	Alkahtani et al. (2020) Anwar and Li (2021), Al Shukaili et al. (2022), and Khan et al. (2020).
	Government credit support plays a role in maintaining stability during crises.	Bertoni et al. (2019), Chundakkadan et al. (2022), and Jiang et al. (2020).
	Government support modifies the relationship between business skills and SME survival and enhances SME performance.	Bongomin et al. (2018), Hansen et al. (2009), Nakku et al. (2020), and Nguyen, Tran, and Do (2018).
	Access to government support is key to fostering innovation and sustainability within SMEs.	Ullah et al. (2021).
	Government support is positive but needs to be improved for greater effectiveness.	Alharbi (2023).
b. Complex for SMEs and Not Effective for SMEs.	Government financial support has indirect effects.	Jayeola et al. (2022), and Vu and Tran (2021).
	The impact of government support varies with firm profitability levels.	Nguyen, Van, et al. (2018).
	Despite the availability of government support, SMEs may resort to self-funding owing to distrust in government initiatives.	Sambajee and Dhomun (2015).
	Government efforts are insufficient in meeting the needs of SMEs, which indicates a common issue of inadequate support.	Anton and Onofrei (2016), Bomani et al. (2022), and Umihanic et al. (2016).
	Government support for SMEs is poorly structured, fragmented, and untargeted, leading to persistent liquidity gaps despite intervention programs.	Klonowski (2010), and Klonowski (2012).
	Financial assistance does not significantly help SMEs.	Doan et al. (2021) and Zulu-Chisanga et al. (2021).
	Certain forms of support, such as subsidies and participative loans, can lead to adverse effects.	Choi and Lee (2020) and Marti and Quas (2018).
C2: The Emergence of Fintech as a Support Mechanism for SMEs	Fintech has a positive impact on SME efficiency and access to finance.	Abbasi, Alam, Du, and Huynh (2021), and Flögel and Beckamp (2020).
	Fintech provides alternatives to traditional banking, especially for those who are financially constrained.	Charaia et al. (2021) and Xiang et al. (2021).
	Fintech played a crucial role for SMEs during COVID-19, supporting business continuity.	Karim et al. (2022) and Y. Zhang et al. (2023).
	Fintech improves SMEs' access to finance and enhances their sustainability and competitiveness.	Pizzi et al. (2021), Siddik et al. (2023), and Soni et al. (2022).
	Convenience and process transparency are pivotal in attracting borrowers to Fintech platforms.	Maier (2016).
	The presence of P2P lending and innovation in financing methods is essential for SME development, with Fintech providing a more accessible financing platform.	Abbasi, Alam, Brohi, et al. (2021), Rosavina et al. (2019), Sybirianska et al. (2018) and X. Zhang et al. (2022).
C3: Government Support in Fostering Fintech Adoption for SMEs	Government plays a role in promoting the use of digital technologies for financial inclusion and supporting Fintech integration.	Feng et al. (2023), Gupta et al. (2022), Huang (2022), Hua and Huang (2021).
	Government support must provide technology training to facilitate SMEs growth and finance access.	Suharyati and Utami (2022).
	Indian government schemes are aimed at enhancing SMEs performance through collaboration with Fintech, and research is needed to assess their impact.	Maiti (2018).
	The UK Bank Referral Scheme is a government initiative that adopts Fintech platforms to aid SMEs in finding alternative financing, yet it achieves only moderate success in changing the established banking structure.	Schammo (2019).
	Government support and regulatory frameworks are important for developing Fintech ecosystems that enhance SME finance.	Semenyuta et al. (2019), and Sheng (2021).
	Hungarian SMEs face challenges in digitalization and innovation, partly due to limited government support.	Endrodi-Kovács and Stukovszky (2022).
	A Nigerian approach combines traditional and innovative financing to meet the varied needs of SMEs.	Lawal et al. (2018).

SME survival. This interaction effect is crucial as it reveals the government's role in not only providing support but also in enhancing SMEs' internal capabilities to effectively utilize this support.

During the COVID-19 pandemic, Chundakkadan et al. (2022) highlighted that government support programs were more accessible to firms that adapted their operations, which suggests that government measures were instrumental for SMEs facing financial constraints. This adaptability is essential for resilience and aligns with the findings from Vietnam, where government assistance significantly influenced SME dynamics (Hansen et al., 2009).

The direct and positive influence of government support on SME performance is also seen in Pakistan's Swat Valley (Khan et al., 2020), reinforcing the notion that state intervention is a key driver of SME development. In South Africa, Msomi and Olarewaju (2021) emphasized the substantial impact of government financial support on SME access to finance, which is essential for their growth and contribution to the economy. Specifically, government financial support programs in South Africa have been tailored to reduce stringent loan access requirements for SMEs, including subsidized loan schemes and government-backed guarantees that lower the perceived risk for financial institutions (Msomi & Olarewaju, 2021). These policies not only increase financial

access but also directly contribute to the survival of SMEs by creating a more inclusive financial environment.

Nakku et al. (2020) further analyzed the interconnectedness of government support programs, entrepreneurial orientation, and SME performance in Uganda. They suggested that government support not only boosts financial performance but also enhances non-financial dimensions, such as entrepreneurial orientation.

This analysis underscores that while traditional government support has been foundational in supporting SME development, these supports need to become more adaptable, responsive, and aligned with modern business needs. The effectiveness of government initiatives, therefore, hinges on their ability to be flexible and integrate deeper insights into the economic landscape, thereby providing not just support but a strategic advantage to SMEs. Moving forward, integrating Fintech solutions could address several of these flexibility and responsiveness challenges, potentially transforming traditional support mechanisms into more effective tools for economic empowerment.

3.2.1.2. Government financial support is complex and ineffective for SMEs. The effectiveness of traditional government support for SMEs is not a straightforward affair as it is laden with complexities that manifest differently across various contexts. As noted by Nguyen, Van, et al. (2018) in Vietnam, the impact of such support on firm profitability is contingent on firms' existing financial health, which suggests that government interventions must be nuanced and adaptable to different levels of profitability within the SME sector.

While the distinction between developing and developed economies highlights varying levels of government intervention, the institutional environment plays a substantial role in determining the success of these programs. In developing economies where institutional frameworks are often weaker, SMEs face additional barriers such as bureaucracy and corruption, which can hinder the full utilization of government support (Bomani et al., 2022; Msomi & Olarewaju, 2021). Conversely, developed economies tend to have more robust institutional structures that enable a quicker and more efficient implementation of government initiatives (Abbasi, Alam, Du, & Huynh, 2021). These challenges highlight the importance of customizing support mechanisms to accommodate SMEs with varying financial health rather than adopting blanket policies that may only benefit financially stronger firms. A more granular, targeted approach could help address these disparities and ensure a more equitable distribution of benefits (Jayeola et al., 2022).

Another critical aspect contributing to the inefficacy of these support mechanisms is the lack of effective communication and outreach. Many SMEs remain unaware of the resources available to them owing to complex application procedures and insufficient government communication strategies (Klonowski, 2010). To enhance the reach and success of these programs, governments need to streamline application processes and improve outreach efforts, ensuring that SMEs know about and can easily access available support programs. Transparency and accessibility are crucial in ensuring that well-intentioned programs effectively reach their intended recipients (Abbasi, Alam, Du, & Huynh, 2021; Bomani et al., 2022).

The effectiveness of government initiatives is further complicated by the level of trust and support perception among businesses. Sambajee and Dhomun (2015) illustrate that in the Maldives and Mauritius, despite better business environments, SMEs may still resort to self-funding strategies owing to a lack of confidence in government initiatives. This trust deficit can significantly limit the effectiveness of even well-intentioned support measures.

Adding another layer to this complexity, Vu and Tran (2021) find, in Vietnam, that government financial support does not directly translate into firm productivity but rather contributes to technological progress and growth on the firm scale. This suggests that the true value of government support may lie in its indirect effects on SME capabilities rather than in immediate productivity enhancements.

In Malaysia, government financial support has shown a limited direct impact on SMEs' financial performance. However, as highlighted by Jayeola et al. (2022), such support plays a crucial role in enabling SMEs to invest in technological advancements, such as ERP (enterprise resource planning) systems and cloud computing solutions, which enhance competitive advantage and long-term operational outcomes. The integration of cloud computing offers SMEs scalable, on-demand resources that were previously accessible mainly to larger firms. This transformation enables SMEs to streamline their operations, improve customer relationship management, and gain access to advanced financial tools without substantial IT investments. Although these upgrades may not yield immediate financial returns, they contribute to building a more resilient and capable SME sector over time. Consequently, evaluating the success of government programs should consider these long-term impacts on organizational capacity and innovation beyond short-term financial performance alone. This strategic approach ensures that government support effectively promotes sustained growth and adaptability within the SME sector.

Despite these intentions, the efficacy of government support programs often falls short of meeting the needs of SMEs, as evidenced by studies across various global contexts. For instance, in Romania, despite the implementation of various public policies and support programs, small businesses still struggle to access finance; this indicates a gap between policy intent and on-the-ground realities (Anton & Onofrei, 2016). A major issue is the lack of coherence and coordination in these support programs, which results in SMEs being either unaware of available resources or unable to access them owing to overly complex procedures (Klonowski, 2010; Umihanic et al., 2016). Streamlining support mechanisms and improving the accessibility and transparency of these programs is essential for increasing their effectiveness. Similar findings were found in Zimbabwe (Bomani et al., 2022), South Korea (Choi & Lee, 2020), Zambia (Zulu-Chisanga et al., 2021), and Vietnam (Doan et al., 2021).

In Poland, Klonowski (2010, 2012) criticized government support programs for being poorly structured and fragmented, which has led to their underutilization and the persistence of liquidity gaps among SMEs. This sentiment is echoed in Bosnia and Herzegovina, where SMEs perceive government support measures as inadequate (Umihanic et al., 2016).

Moreover, Bomani et al. (2022) found that urban SMEs struggle with accessing government support owing to high interest rates and stringent collateral requirements. Even SMEs close to subsidy eligibility—particularly those with lower profitability and innovation—face the "Peter Pan syndrome," remaining dependent on subsidies without substantial growth (Choi & Lee, 2020). This reflects the importance of addressing structural issues within these programs to prevent SMEs from becoming reliant on subsidies while failing to grow. Government support should focus on helping SMEs transition from dependency to sustainability. The disconnection between financial aid and its intended outcomes of improving business performance appears to persist even when coupled with technical support (Doan et al., 2021).

In Spain, receiving participative loans results in significantly higher external financial debts, which raises questions about the sustainability of such funding mechanisms (Marti & Quas, 2018). Evidence has revealed that instead of government support, factors such as inter-firm collaboration and managerial ties play a more definitive role, underscoring the importance of relational and managerial dynamics over direct financial support (Zulu-Chisanga et al., 2021).

Addressing RQ1, these findings highlight the urgent need for government support systems to evolve beyond traditional frameworks. This evolution should incorporate greater flexibility, better alignment with technological advancements, and deeper insights into the unique challenges faced by SMEs. By understanding these challenges, policymakers can design interventions that truly provide support.

3.2.2. Sector 2: The emergence of fintech as a support mechanism for SMEs

This sector brings a critical focus to Fintech's emerging role in SME financing, addressing RQ1 by identifying where it can serve as an alternative to traditional government support and the specific challenges it faces. Simultaneously, it informs RQ2 by showcasing how Fintech solutions could potentially integrate with government strategies to create a more effective financial ecosystem for SMEs.

As traditional government support for SMEs shows a spectrum of effectiveness and complexity, Fintech emerges as a transformative force, charting a new course for SME financial accessibility and efficiency. Studies such as those by Abbasi, Alam, Brohi, et al. (2021) and Abbasi, Alam, Du, and Huynh (2021) indicate a direct positive impact of Fintech on SME efficiency and loan access within OECD countries, particularly through P2P lending platforms. This underscores the growing significance of Fintech in enhancing SME financial operations, challenging traditional credit institutions, and providing viable alternatives where traditional approaches fall short.

However, Fintech's role should not be viewed as a total replacement for traditional banking institutions. Rather, in regions such as Germany, Fintech has demonstrated potential as a complementary mechanism to traditional banking, particularly in overcoming informational asymmetries (Flögel & Beckamp, 2020). The collaboration between Fintech and regional banks could foster a hybrid financial model, strengthening SME access to finance through combined innovative solutions and established trust.

Charaia et al. (2021) noted that in regions such as Georgia, Azerbaijan, and Armenia, while traditional credit-providing institutions constrain the SME sector, Fintech offers progressive solutions; this signals a shift in the financial paradigm. Similarly, in Indonesia, SMEs are increasingly turning to P2P lending platforms owing to the limitations of traditional finance systems, which often impose stringent collateral requirements (Rosavina et al., 2019).

These platforms offer more flexible solutions, which makes them a viable alternative for SMEs that struggle to access conventional financial institutions.

The COVID-19 pandemic has further underscored the importance of Fintech in ensuring SME resilience. Karim et al. (2022) observed the crucial "collisions" between Fintech growth and SME strength, particularly in ASEAN countries, where conventional financial avenues were disrupted. This highlights Fintech's agility in crisis management, which has become essential for the survival and growth of SMEs. The pandemic acted as a catalyst, fast-tracking the digital adoption of financial tools, and encouraging SMEs to seek out Fintech platforms as lifelines during economic turbulence.

The narrative of Fintech as a facilitator for SME development was further enriched by another study (Maier, 2016) emphasizing that factors such as convenience and process transparency on crowdlending platforms can be more influential than economic criteria in attracting borrowers. This reflects a broader trend where Fintech is revolutionizing access to finance for SMEs, fostering an environment that supports innovation and growth.

In the face of traditional finance constraints, SMEs in regions such as China increasingly turn to Fintech as an essential channel for financial access (Xiang et al., 2021). Further, Fintech has been more effective than traditional finance in shielding SMEs from cash crunches during the pandemic and in supporting recovery efforts (Y. Zhang et al., 2023).

Fintech innovations such as invoice factoring, digital payments, and blockchain technologies are expanding the range of tools available to SMEs, facilitating access to credit while also enhancing operational efficiency and financial transparency (Feng et al., 2023; Gupta et al., 2022). Blockchain, in particular, increases trust by providing an immutable ledger; for example, Russia's unified depositor registry leverages blockchain to ensure data integrity and transparency (Semenyuta et al., 2019). This immutability makes blockchain an effective tool for building trust in financial transactions (Semenyuta et al., 2019). Furthermore, blockchain's potential applications in real-time payment settlements and cross-border transactions could reduce reliance on intermediaries, lower transaction costs, and benefit SMEs in global supply chains (Pizzi et al., 2021; X. Zhang et al., 2022).

As Fintech platforms increasingly handle sensitive financial data, data privacy and cybersecurity concerns grow, especially in regions with limited cybersecurity infrastructure. Studies have emphasized that regulatory gaps can lead to data abuse and insufficient privacy protections, as seen in China's rapidly expanding Fintech sector, where information monopolies and privacy vulnerabilities are notable risks (Hua & Huang, 2021).

Addressing these concerns is essential for building SME trust in Fintech solutions, especially in regions with developing financial systems, where limited security measures could hinder adoption. Ensuring robust technological frameworks and supportive regulatory measures will be critical to overcoming these challenges (X. Zhang et al., 2022).

Collectively, these studies portray Fintech not only as a financial tool but also as a strategic partner for SMEs, providing innovative avenues for financing that align with the technological evolution and dynamic market demands of the modern era. They suggest a future where Fintech can bridge gaps left by traditional financial systems, potentially becoming integral to SME success stories. This potential is especially prominent in the context of Industry 4.0, where Fintech's role in supply chain efficiency and sustainable business practices positions it as a key player in the next wave of industrial advancement (Soni et al., 2022).

Despite the promising outlook, Fintech adoption does face challenges; one of these is uneven development across regions, such as the underdeveloped Fintech landscape in Ukraine (Sybirianska et al., 2018). In addition to infrastructure challenges, regulatory ambiguity remains a substantial hurdle. In many developing regions, unclear or inconsistent regulations around Fintech solutions create confusion for both providers and SMEs, slowing down adoption (Choi & Lee, 2020). Additionally, trust-building between SMEs and Fintech platforms is crucial, particularly in regions where unfamiliarity with digital financial solutions can heighten skepticism (Gupta et al., 2022; Rosavina et al., 2019). Concerns over perceived risks highlight the need for enhanced trust and regulatory measures to mitigate potential negative effects and ensure the equitable advancement of Fintech solutions globally.

The full realization of Fintech's potential is thus contingent upon addressing these challenges, ensuring that Fintech solutions are well tailored to meet the diverse needs of SMEs across different contexts. The integration of Fintech in SME financing strategies represents a promising area—albeit complicated—in the effort to support SME growth and resilience. Establishing clear and stable regulatory frameworks is essential for governments to foster confidence in the system and encourage SMEs to utilize Fintech platforms (Hua & Huang, 2021).

3.2.3. Sector 3: Government support in fostering fintech adoption for SMEs

This sector addresses RQ2 by exploring how government initiatives facilitate Fintech adoption in the SME sector and the resulting impact on SME financial access, growth, and overall performance. It highlights how these efforts create enabling environments that contribute to the financial empowerment of SMEs, thus promoting broader economic resilience.

In Hungary, initiatives reveal a pressing need for enhanced digitalization efforts to close gaps with more technologically advanced regions of the EU (Endrodi-Kovács & Stukovszky, 2022). This anticipatory approach underpins the necessity for governments in catching-up economies to adopt aggressive digital strategies, preparing the SME sector for future technological integration.

In China and Russia, governments actively drive the implementation of Fintech solutions. Strategic regulatory frameworks and policy initiatives are developed to foster innovation while ensuring stability and consumer protection (Semenyuta et al., 2019; Sheng, 2021). For instance, Russia's strategic move toward digitalization in SME lending focuses on how government-supported digital technologies can surpass established European models in accessibility and efficiency (Semenyuta

et al., 2019).

The regulatory efforts in China are pivotal, creating a financially tolerant environment and market shortages (Hua & Huang, 2021; Huang, 2022). These policies facilitate deeper integration into SME finance, enhancing Fintech's role in closing financial access gaps and promoting sustainable economic growth.

Moreover, in emerging economies, the findings underscore the pivotal role of digital finance, where effective government intervention can significantly enhance the accessibility of financial services to SMEs (Feng et al., 2023; Hua & Huang, 2021). These studies have advocated a collaborative approach to policymaking, emphasizing the creation of supportive environments and ensuring that Fintech solutions are tailored to the specific needs of SMEs, thereby promoting sustainable economic growth. Continuously assessing and optimizing these policies to support SMEs' growth and adapt to evolving market conditions is essential.

In India, government initiatives aimed at enhancing SMEs' capabilities through strategic partnerships with Fintech firms have been highlighted as effective (Maiti, 2018). These policies are deliberately designed to leverage Fintech innovations, directly contributing to the economic empowerment of SMEs. Conversely, in the UK, the Bank Referral Scheme, intended to facilitate such partnerships, has achieved only moderate success (Schammo, 2019). This scheme's limited impact underscores the challenges inherent in integrating Fintech solutions within traditional financial systems that have not undergone substantial systemic reforms.

Similarly, Lawal et al. (2018) described the Nigerian government's shift beyond traditional support mechanisms by fostering innovative partnerships with Fintech companies—a transformative approach involving direct partnerships aimed at rapid technological advancement. These differences underscore the necessity for adaptability and innovation in environments with less developed financial infrastructures, highlighting how tailored government interventions must be to effectively support Fintech adoption depending on each country's economic conditions.

Moreover, Indonesia exemplifies a comprehensive approach to enhancing both the infrastructure and human capital necessary for Fintech adoption. Suharyati and Utami (2022) provided an in-depth look at how the Indonesian government actively improves the quality of Fintech services and invests in essential training for SMEs. Government policies supporting digital financial literacy are equally crucial. Training programs and educational initiatives should be integrated into government strategies to ensure that SMEs can effectively leverage Fintech for growth and operational efficiency. This strategy ensures that the workforce is adequately prepared to utilize advanced financial tools; this highlights the critical importance of aligning technological upgrades with workforce development to maximize the potential for economic growth and financial inclusion.

By weaving together these diverse international perspectives, we observe that the effectiveness of government initiatives in fostering Fintech adoption hinges on their ability to be flexible, responsive, and integrative. Governments are not just facilitators of technological adoption but also strategic architects, crucially shaping the ecosystems in which these technologies operate. This comprehensive approach ensures that policies are not only supportive but transformative, enhancing SME financial access and promoting sustainable economic growth.

Further opportunities lie in cross-border Fintech collaboration. Regional initiatives such as ASEAN's digital integration are critical examples of how coordinated governmental efforts can allow Fintech to scale, benefiting SMEs engaged in international trade. Harmonizing regulations across borders would enable Fintech companies to expand their offerings globally, unlocking greater access to finance for SMEs (Feng et al., 2023).

The successful integration of Fintech innovations into SME financing strategies hinges on continued government collaboration and education, alongside a commitment to developing a regulatory framework that promotes equitable and sustainable growth in the Fintech sector. Only through such comprehensive and integrated approaches can governments fully leverage the potential of Fintech to empower SMEs financially, ensuring that the advancements in technology translate into real economic benefits for small and medium enterprises.

The analysis in Table 3 illustrates a comprehensive evolution of government support programs for SMEs from 2009 to 2023, highlighting an ongoing shift in policy focus. These programs have transitioned from traditional financial support mechanisms toward more innovative, technology-enabled initiatives aimed at addressing both financial and operational barriers. Recent trends have emphasized digital finance and Fintech collaborations, reflecting an adaptive response to the growing role of digital technologies in SME growth and sustainability. This progression indicates an increased governmental commitment to creating resilient, inclusive support ecosystems that align with SMEs' evolving needs in a digitalized global economy. Such developments provide a reference for policymakers seeking to design future interventions that effectively integrate traditional support structures with modern digital tools, thereby offering scalable solutions to meet the complexities of contemporary SME challenges.

4. Future research directions

Building on the insights from our bibliometric analysis and addressing RQ3, this section outlines potential research avenues that explore the evolving role of government support and Fintech innovations in enhancing SME financial access. These directions aim to address identified gaps, encouraging deeper exploration of how collaborative efforts can effectively meet the diverse financial needs of SMEs.

4.1. Cross-national comparative studies

Future research should prioritize cross-national comparative studies that examine how different governments' support for SMEs and Fintech innovations affect access to finance. For instance, research by Hua and Huang (2021) demonstrated China's rapid adoption of Fintech due to government policies, while Gupta et al. (2022) highlighted India's strategic use of Fintech within government-backed frameworks such as the Digital India initiative. Future studies should explore these differing national approaches in greater depth to derive policy lessons applicable to other regions (Abbasi, Alam, Du, & Huynh, 2021). This comparison is crucial to uncovering best practices for integrating government support and Fintech, especially in emerging markets where digitalization gaps, as seen in Hungary (Endrodi-Kovács & Stukovszky, 2022), remain a challenge.

4.2. Expanding methodologies and integrating technological insights

As Fintech evolves, research should adopt innovative methodologies, including blockchain analysis, digitalization, and advanced econometric models, to better understand their impact on SME financial access (Feng et al., 2023). Emerging themes such as "blockchain" and "digitalization" indicate growing relevance in the Fintech landscape. Future research should incorporate these advanced technologies to explore their role in improving SME financial access. This will provide a deeper understanding of how these tools enhance transaction efficiency and credit evaluation, leading to more precise policy interventions such as developing regulatory frameworks and promoting educational initiatives.

4.3. Sectoral analysis and influential variables

Researchers should explore how factors such as business size, industry sector, and the adoption of digital tools affect SME access to finance through Fintech (Maiti, 2018). For instance, a study on Germany's Fintech ecosystem by Flögel and Beckamp (2020) showed how Fintech collaborates with traditional banks, revealing sector-specific

Table 3
Timeline of government support programs for SMEs (2009–2023)

Year	Country	Government Program/Objective	Key Impact on SME	Success Measure	Reference
2009	Vietnam	Government assistance for SME dynamics	Supports SME dynamics	Effective	Hansen et al. (2009)
2010	Poland	Financial assistance programs	Poor program structure	Not Effective	Klonowski (2010)
2012	Slovakia & Poland	Liquidity intervention programs	Persistent liquidity gaps	Complex	Klonowski (2012)
2015	Maldives & Mauritius	Financial access & business support	Limited effectiveness in Maldives; Distrust in Mauritius	Complex	Sambajee and Dhomun (2015)
2016	Romania	Various public policies and support programs	Limited finance access	Not Effective	Anton and Onofrei (2016)
2016	Bosnia and Herzegovina	Government grants	Inadequate grants	Not Effective	Umihanic et al. (2016)
2018	Nigeria	Bank of Industry, credit schemes, agricultural funds	Enhanced credit access	Effective	Lawal et al. (2018)
2018	Vietnam	Government support for urban SMEs	Varies with profitability	Complex	Nguyen, Van, et al. (2018)
2018	Vietnam	Tax breaks and loans for financial performance	Enhances financial performance	Effective	Nguyen, Tran, and Do (2018)
2018	Uganda	Government support moderating SME survival factors	Supports survival	Effective	Bongomin et al. (2018)
2018	Spain	Participative loans for SMEs	Increases financial debt	Not effective	Marti and Quas (2018)
2018	India	Bank initiatives via Fintech collaboration	Enhanced finance access	Effective	Maiti (2018)
2019	Spain	ENISA participative loans	Improves growth	Effective	Bertoni et al. (2019)
2019	Russia	Digital technologies in SME lending	Enhanced accessibility	Effective	Semenyuta et al. (2019)
2019	UK	Bank Referral Scheme for SME financing	Limited success	Complex	Schammo (2019)
2020	Pakistan	Financial support for competitive performance	Increases competitive performance	Effective	Alkahtani et al. (2020)
2020	Pakistan	Government incentives, credit schemes for competitiveness	Enhances competitiveness	Effective	Khan et al. (2020)
2020	China	Government credit support during COVID-19	Supports financial stability	Effective	Jiang et al. (2020)
2020	South Korea	Government subsidies impact on SMEs	Dependency challenges	Complex	Choi and Lee (2020)
2020	Uganda	Financial and non-financial support programs for SMEs	Enhances EO impact on performance	Effective	Nakku et al. (2020)
2021	China	Interest rate subsidies for SME loans	Increases loan access	Effective	Wang et al. (2021)
2021	China	Regulatory tolerance in Fintech	Improved investment efficiency	Effective	Hua and Huang (2021)
2021	Vietnam	Financial and technical assistance for SME efficiency	Limited efficiency impact	Not Effective	Doan et al. (2021)
2021	China	Government support for Fintech-driven efficiency	Enhances credit efficiency	Effective	Sheng (2021)
2021	Zambia	Support systems for SMEs	No impact on performance	Not Effective	Zulu-Chisanga et al. (2021)
2021	South Africa	Loan access reduction policies	Facilitates growth	Effective	Msomi and Olarewaju (2021)
2021	Vietnam	Financial support to enhance technological progress and scale efficiency	Advances technological progress, with limited productivity impact	Complex	Vu and Tran (2021)
2021	China	Government incentives for SME competitiveness	Effective credit schemes	Effective	Anwar and Li (2021)
2021	Pakistan	Financing programs for sustainability	Enhances sustainability	Effective	Ullah et al. (2021)
2022	Hungary	Limited support for digitalization	Lagging digitalization	Complex	Endrodi-Kovács and Stukovszky (2022)
2022	Malaysia	Cloud ERP for competitive advantage through GFS	Enhances competitive advantage	Effective	Jayeola et al. (2022)
2022	India	Fintech support for operational efficiency	Improves transaction efficiency	Effective	Gupta et al. (2022)
2022	Indonesia	Tech training for MSMEs	Improves loan access	Effective	Suharyati and Utami (2022)
2022	Oman	Government-backed firm support	Enhances employment	Effective	Al Shukaili et al. (2022)
2022	Saudi Arabia	Stimulus program improvements	Need for tech upgrades by government	Not Effective	Alharbi (2023)
2022	Zimbabwe	Government support for post-war SME survival	Insufficient support	Not Effective	Bomani et al. (2022)
2022	China	Government policies supporting Fintech for SME financing	Improved investment efficiency	Effective	Huang (2022)
2022	Indonesia	Loans, subsidies, and credit restructuring programs	Improves loan accessibility	Effective	Supari and Anton (2022)
2022	India	Targeted government support for financially constrained firms	Stabilizes firms	Effective	Chundakkadan et al. (2022)
2023	China	Regulatory measures supporting digital finance	Reduces financing constraints & encourages sustainability	Effective	Feng et al. (2023)

challenges. Examining such variables across different regions—particularly in emerging markets (Suharyati & Utami, 2022)—can uncover how Fintech solutions can be tailored to meet the unique needs of each sector and region.

4.4. Cybersecurity and regulatory adaptability

With the increasing reliance on digital financial tools, cybersecurity concerns are paramount, particularly for SMEs that often lack the resources to implement robust security measures. As noted in our publication trends, cybersecurity emerged as a major concern during the COVID-19 pandemic, indicating the need for stronger regulatory frameworks. Future research should focus on how Fintech providers and policymakers can collaborate to create secure, adaptable digital ecosystems (Javaheri et al., 2024; Suharyati & Utami, 2022). Additionally, exploring how regulatory frameworks can evolve to balance innovation with security and consumer protection is essential, especially in regions where these frameworks are adapting to new technological advancements (Hanafizadeh & Gerami Amin, 2023; Hua & Huang, 2021).

4.5. Leveraging big data analytics for government-supported fintech solutions

Future research should explore how government-backed Fintech platforms can leverage big data analytics to further enhance SME access to finance. By integrating big data into Fintech solutions, governments can improve the accuracy of credit evaluations, personalize financial

products for SMEs, and predict financial needs more effectively. This approach can enable more efficient deployment of financial resources, especially in underserved regions. Acciarini et al. (2023) emphasized the potential of big data to support business model innovation by enabling better decision-making and customer insights. Additionally, Zhou and Chen (2021) highlighted the role of reputation mechanisms in balancing Fintech innovation and regulation. Incorporating these technologies into government-supported Fintech platforms could drive more impactful and tailored financial interventions for SMEs, fostering a more inclusive financial ecosystem.

5. Policy recommendations

Addressing RQ3, this section outlines policy interventions derived from our empirical insights, focusing on how government initiatives and Fintech solutions can synergistically enhance SME financial access. These recommendations are aimed at guiding policymakers in strengthening financial ecosystems, promoting Fintech adoption, and improving SME resilience and growth.

5.1. Strengthening financial ecosystems through collaboration

Governments should facilitate collaborations between Fintech platforms and traditional banks to improve SME financing. For example, our collaboration network analysis highlighted the importance of partnerships between financial institutions and digital platforms in countries such as China and Germany (Flögel & Beckamp, 2020; Hua & Huang, 2021). Such collaborations bridge the gap between conventional finance and innovative digital solutions, ensuring SMEs have access to diverse financing options.

5.2. Tailoring legislative frameworks to promote fintech innovation

To support Fintech growth, governments should continuously update digital infrastructure and regulatory frameworks that balance innovation with investor protection. Countries such as Singapore and the UK offer models of regulatory sandboxes that promote controlled innovation (Charaia et al., 2021). These frameworks should also be adapted to meet the specific challenges of emerging markets, ensuring they support SME needs in different economic contexts.

5.3. Evaluating government support mechanisms in crisis situations

Government support mechanisms, such as Partial Credit Guarantee Schemes and crowdfunding platforms, should be evaluated for their effectiveness in times of crisis. The challenges faced by SMEs during COVID-19 underscore the importance of having robust systems in place to ensure SMEs can access capital during economic downturns (Vu & Tran, 2021).

5.4. Promoting sustainable and technologically advanced financial solutions

Fintech can play a key role in promoting sustainable SME growth by supporting the adoption of Industry 4.0 business models. Policymakers should incentivize the use of Fintech for resource-efficient practices, especially in regions where conventional financial systems are underdeveloped (Pizzi et al., 2021). This will help SMEs transition to sustainable and resilient business models, ensuring long-term financial stability.

5.5. Public-private partnerships and digital financial literacy

Governments should establish public-private partnerships with Fintech firms to co-develop solutions aimed at improving financial inclusion, particularly in underserved regions. The example of Indonesia's collaboration with P2P lending platforms, alongside government regulations by the Financial Services Authority of Indonesia (OJK), demonstrates how government facilitation can lead to the increased adoption of Fintech solutions, expanding SME access to financial services (Rosavina et al., 2019). Additionally, the Indonesian government's support for digital financial literacy programs, coupled with technology training initiatives for SMEs, highlights the importance of ensuring that SMEs can effectively leverage Fintech tools for growth and operational efficiency (Suharyati & Utami, 2022).

6. Conclusion

In this SLR, we adhered to PRISMA guidelines to ensure a structured and transparent approach, systematically analyzing the impact of government policies and Fintech innovations on SME financial access. Utilizing comprehensive search strategies across Scopus and Web of Science, we focused on articles published from 2009 onward, which culminated in a dataset of 54 relevant studies. Employing both bibliometric and qualitative content analyses, our methodology allowed for a nuanced exploration of empirical data and discussions, bridging gaps and highlighting areas for further exploration.

Addressing the first research question (RQ1), which explores the effectiveness of government support and Fintech in enhancing SME access to finance when considered separately, we identified substantial gaps and challenges in existing frameworks. This highlighted a critical need for regulatory frameworks that are nuanced and aligned with financial inclusion and SME empowerment objectives (Hua & Huang, 2021). Our results revealed that despite the presence of government-backed initiatives, such as guaranteed loan schemes and subsidized loans, the effectiveness of these measures is often constrained by bureaucratic inefficiencies and lack of accessibility, particularly in emerging economies. The complexities and limitations of traditional government support for SMEs were evident, underscoring the importance of strategic interventions and regulatory adaptability to foster a more inclusive financial ecosystem (Umihanic et al., 2016).

While Fintech innovations have shown considerable promise in bridging the financial access gap for SMEs—especially through digital financial services, adoption of technology, and alternative lending platforms—their impact remains inconsistent across different regions owing to varying levels of digital infrastructure and regulatory support (Hua & Huang, 2021; Huang, 2022). In emerging economies, where traditional financial systems often fail to adequately meet the needs of SMEs, Fintech solutions can play a transformative role by providing more accessible, efficient, and tailored financial services (Siddik et al., 2023). However, this potential is frequently hindered by a complex interplay of regulatory barriers, limited digital literacy, and insufficient technological infrastructure (Feng et al., 2023; Huang, 2022).

In reviewing the impact of government policies on SME financial access, recognizing potential unintended consequences is also crucial. While government policies such as digital infrastructure support have facilitated SME access to finance, certain fiscal measures can have adverse effects. For instance, research has indicated that electronic levies on mobile money transactions can inadvertently restrict SMEs' access to vital digital financial services (Pobee et al., 2023). This restriction may lead to reduced usage and adoption of these services, posing considerable challenges to financial inclusion in developing economies. Such findings emphasize the need for policymakers to carefully consider the broader economic impacts of fiscal policies. Ensuring that these policies do not inadvertently hinder the financial empowerment of SMEs but rather support their growth and integration into the financial system is essential.

Delving into the second research question (RQ2), our findings illuminated the multifaceted roles of government support and Fintech interventions. We demonstrated their combined potential not only to enhance SMEs' access to finance but also to help them navigate and overcome traditional financial barriers, marking a pivotal step toward

inclusivity in SME financing.

For example, Fintech innovations, such as alternative lending platforms and mobile banking, have successfully complemented government-backed financial schemes in China, alleviating financing constraints and enhancing SME investment efficiency (Huang, 2022). Similarly, Hua and Huang (2021) presented successful cases where government policies, such as digital infrastructure support and regulatory sandboxes, have facilitated the growth of Fintech ecosystems, thereby empowering SMEs with more diverse financing options.

However, despite these promising examples, our analysis also identified several challenges that need to be addressed to optimize the synergy between government support and Fintech innovations. These include regulatory fragmentation, inconsistent policy implementation, and varying levels of technological readiness across different regions (Karim et al., 2022). Addressing these issues will require coordinated efforts between governments, Fintech providers, and other stakeholders to create a cohesive and supportive ecosystem.

Furthermore, the most effective interventions were those that tailored Fintech solutions to the specific financial needs of SMEs, taking into account regional economic contexts and industry-specific challenges (Pizzi et al., 2021; Siddik et al., 2023; Xiang et al., 2021). This underscores the importance of developing customized strategies that integrate Fintech with traditional financial systems to maximize the benefits for SMEs. Collaborative efforts, such as public–private partnerships and the establishment of regulatory sandboxes, have proven effective in several case studies across Asia and Europe; this highlights the potential for these approaches to be replicated in other regions (Pizzi et al., 2021; Siddik et al., 2023).

Addressing the third research question (RQ3), this review underscores the importance of advancing cross-national understanding and policy efficacy. Future research should embark on comparative studies to explore the nuanced impacts of government and Fintech initiatives on SME financial empowerment, building upon insights provided by Feng et al. (2023) and Pizzi et al. (2021). Such studies are crucial for expanding the knowledge base, refining policymaking, and optimizing strategic government interventions through Fintech collaborations to contribute to the sustainable development of SMEs globally.

In concluding our SLR, we recognize certain limitations. Specifically, the literature reflecting more focus on developed Fintech ecosystems which may limit the generalizability of our findings to less digitally advanced regions. This underscores the need for future research to broaden its geographical focus—particularly toward emerging economies—to enhance the global applicability and relevance of our findings. Expanding both the geographical and methodological scope will provide a more holistic understanding of the challenges and opportunities facing SMEs in diverse contexts.

While our review included studies addressing both government support and Fintech contributions to SME financing, covering publications up to the current year, it may not capture the most recent shifts or emerging trends in policy or technological advancement post-publication. Future research should aim to continuously update these findings to reflect ongoing developments, providing a dynamic understanding of the evolving landscape of SME financing.

This landscape, underscored by technological advancements and emerging challenges such as cybersecurity and the need for regulatory adaptability, necessitates a proactive approach from SMEs, policymakers, and Fintech companies (Hanafizadeh & Gerami Amin, 2023; Javaheri et al., 2024). These stakeholders must collaborate effectively to adapt to ongoing changes, particularly in areas where technological innovations vary significantly by region (Sanga & Aziakpono, 2023). For research to remain relevant across diverse economic contexts, it must consider localized impacts, especially in emerging markets with varying regulatory frameworks and infrastructure.

Moreover, the practical implications of successful Fintech initiatives, such as eTradeConnect in Hong Kong, highlight the transformative potential of Fintech solutions for SME financing. Supported by the Hong Kong Monetary Authority, this development demonstrates the role of government in enabling technological innovations that enhance trade finance efficiency (HKMA, 2023). This reflects the findings of Abbasi, Alam, Brohi, et al. (2021), who emphasized the integration of government-backed Fintech ecosystems to enhance SME access to finance. Similarly, efforts such as Project Greenprint by the Monetary Authority of Singapore (Monetary Authority of Singapore, 2023) and the Fintech Hive by Dubai International Financial Centre (DIFC, 2023) showcase the effectiveness of supportive regulatory environments in fostering Fintech growth and innovation. These efforts echo findings from Pizzi et al. (2021) and Y. Zhang et al. (2023), who highlighted the synergy between government support and Fintech solutions in promoting SME financial resilience during times of crisis, such as the COVID-19 pandemic. Such partnerships are crucial not only for fostering innovation but also for ensuring financial stability and customer trust in the digital age (Abbasi, Alam, Du, & Huynh, 2021).

These examples affirm the critical role of government and regulatory frameworks in fostering financial inclusion and empowering SMEs through technological advancements, advocating for a synergistic approach where government support is integrated with innovative Fintech solutions to maximize SME financial empowerment and economic resilience.

Our review contributes to the academic dialogue by mapping out the current state and potential future of SME financing, offering a foundation for ongoing research and policy development aimed at enhancing SME financial access and empowerment in the digital age. By leveraging these insights, the broader research and policy community can effectively respond to and capitalize on the opportunities presented by the evolving landscape of SME financing.

CRediT authorship contribution statement

Najla Abu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Filipa Pires da Silva:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Pedro Rino Vieira:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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